**AIN SHAMS UNIVERSITY**
**FACULTY OF ENGINEERING**
*CREDIT HOURS ENGINEERING PROGRAMS*
*All Programs (Freshmen)*

**Major Task (Part 2)**

**PHM112: Mathematics (2)**

15

**Course Coordinator: Dr. Makram Roshdy**

**Name:** Mahmoud Hassan **ID:** 23P0104. **Deadline: Week**

*Please, Solve each problem in its assigned place ONLY (the empty space below it)*

## A. ITERATED INTEGRALS

i) Evaluate $\int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \int_{-1}^{5} \cos y \; dx \, dy$

$\cancel{x \cos y \Big|_{-1}^{5}}$

$$\int_{-1}^{5} dx \int_{\pi/6}^{\pi/3} \cos y \; dy$$

$$\cancel{6 \times 0.087} \quad 6 \times \frac{1}{2} = 3$$

ii) Find the volume of the solid that lies under the plane 3x + 2y + z = 12 and above the rectangle $R = \{(x, y) \mid 0 \le x \le 1, -2 \le y \le 3\}$.

$$z = -3x - 2y + 12$$

$$\int_{-2}^{3} \int_{0}^{1} -3x - 2y + 12 \quad dx \, dy$$

$$-\frac{3}{2}x^2 - 2yx + 12x \Big|_{0}^{1}$$

$$\int_{-2}^{3} -\frac{3}{2} - 2y + 12 \quad dy$$

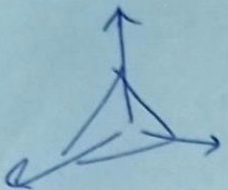
$$-\frac{3}{2}y - y^2 + 12y \Big|_{-2}^{3} = 47.$$

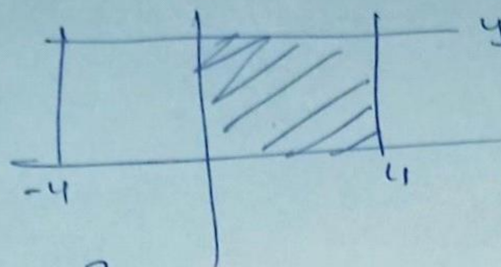
Name: Mahmoud Hassan ID: 23P0104. 

iii) Find the volume of the solid in the first octant bounded by the cylinder $z = 16 - x^2$ and the plane $y = 5$.

$z = 0$ $\quad x^2 = 16$

$y = 5$ $\quad x = -4$



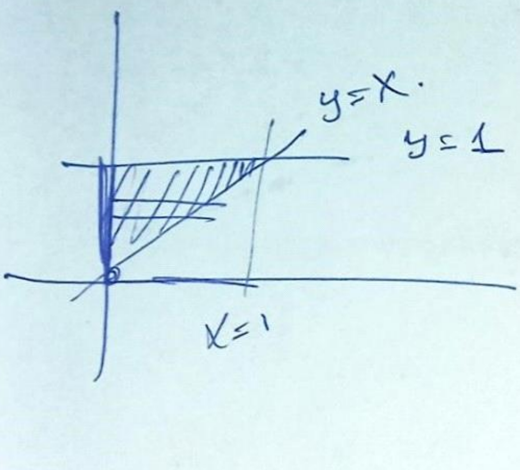



$$\int_0^5 \int_0^4 16 - x^2 \, dx \, dy \rightarrow$$

$$5 \cdot \frac{128}{3}$$

## B. DOUBLE INTEGRALS OVER GENERAL REGIONS

i) Find the volume of the solid enclosed by the paraboloid $z = x^2 + 3y^2$ and the planes $x = 0, y = 1, y = x, z = 0$.




$$\int_0^1 \int_0^y x^2 + 3y^2 \, dx \, dy$$

$$\int_0^1 \frac{x^3}{3} + 3y^2 x \Big|_0^y \, dy$$

$$\frac{y^3}{3} + 3y^3.$$

$$\int_0^1 \frac{10}{3} y^3 \rightarrow \frac{10}{12} y^4 \Big|_0^1$$

$$\frac{10}{12}$$

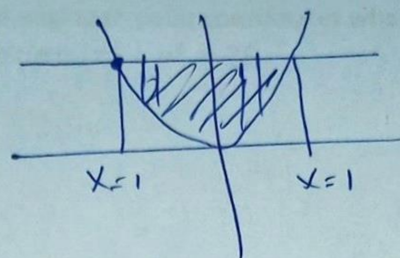
**Mathematics (2)**

**Name:** Mahmoud Hassan **ID:** 23P0104. **3/12**

ii) Find the volume of the solid enclosed by the parabolic cylinder $y = x^2$ and the planes $z = 3y$, $z = 2 + y$.

$2y - 2$

$z = 3y = z = 2 + y$



$$\int_{-1}^{1}\int_{x^2}^{1} 3y - 2 - y \; dy \, dx$$

$y - 1$

$3y = 2 + y$

$2y = 2$

$y = 1$

$$2\int_{-1}^{1} \frac{y^2}{2} - y \Big|$$

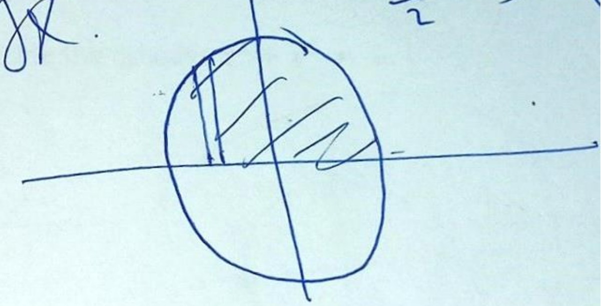
$x = \pm 1$

$$2 \int_{-1}^{1} \frac{x^4}{2} - x^2 + \frac{1}{2} \Big| \rightarrow \int_{-1}^{1} x^4 - 2x^2 + 1 \; dx =$$

iii) Sketch the region of integration : $\int_0^3 \int_{-\sqrt{9-y^2}}^{\sqrt{9-y^2}} f(x, y)\, dx\, dy$ hence change the order.

$x =$

$$\int_{-3}^{3}\int_{0}^{\sqrt{9-x^2}} f(x, y)\, dy\, dx$$



iv) Evaluate the integral by reversing the order of integration $\int_0^8 \int_{\sqrt[3]{y}}^{2} e^{x^4}\, dx\, dy$

$x = \sqrt[3]{}$

$y =$

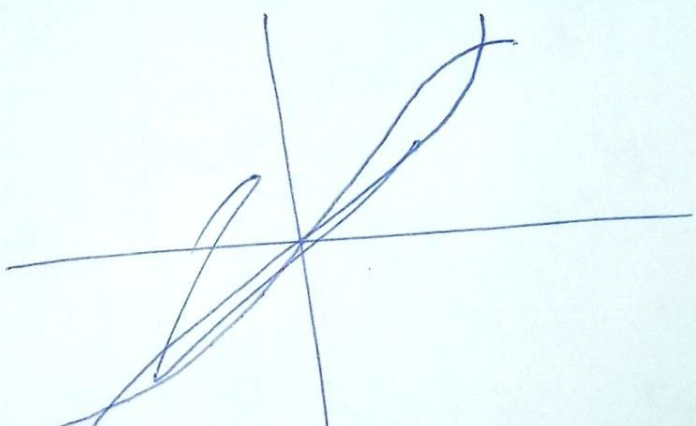
$$\int_0^2\int_0^{x^3} e^{x^4}\, dy\, dx$$

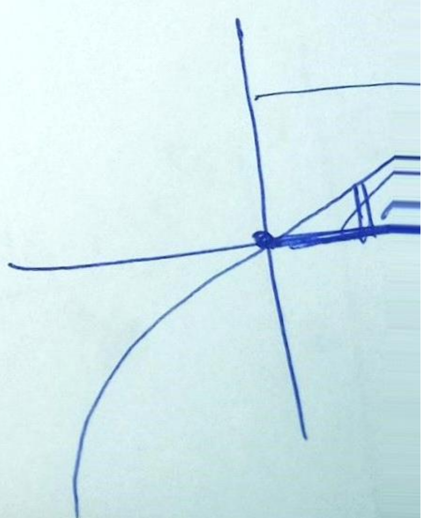
$$\int_0^2 y e^{x^4} \Big|_0^{x^3} dx$$

$$\int_0^2 x^3 e^{x^4}$$

$$\frac{1}{4}\left(e^{x^4}\right)\Big|_0^2$$

$$\frac{1}{4}\left(e^{16} - 1\right).$$





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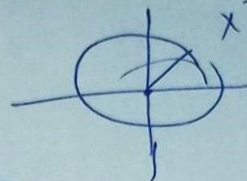
## C. DOUBLE INTEGRALS IN POLAR COORDINATES

i) Evaluate $\iint_R ye^x \, dA$ by changing to polar coordinates where R is the region in the first quadrant enclosed by the circle $x^2 + y^2 = 25$.

$x^2+y^2=5$



$\iint_R ye^x \, dA$

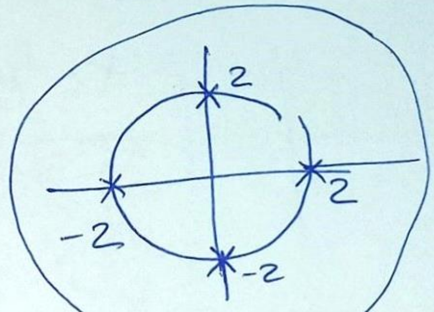
$\int_0^{2\pi}\int_0^{5} r^2 \sin\theta \, e^{r\cos\theta} \, dr \, d\theta$

$\rightarrow \int_0^{5}\int_0^{2\pi} r^2 e^{r\cos\theta} \sin\theta \, d\theta \, dr$

ii) Use Polar coordinates to find the volume of the given solid.

1. Inside the sphere $x^2 + y^2 + z^2 = 16$ and outside the cylinder $x^2 + y^2 = 4$.

$\iint \cancel{\sqrt{16-x^2-y^2} - x^2+y^2}$



$z^2 = 16-x^2-y^2$

$z = \sqrt{16-x^2-y^2}$

$\sqrt{16-(x^2+y^2)}$

$16-(r)^2$

$\int \sqrt{16-x^2-y^2}$

$\int_0^{2\pi}\int_2^{4} (16-r^2)^{1/2} \, r \cdot dr \, d\theta$

$2r$

$\frac{1}{3}\left.\frac{(16-r^2)^{3/2}}{\cancel{3/2}}\right|_2^4$

$\frac{1}{3} \cdot 2\pi \cdot$

$87$

$\int_0^{2\pi} d\theta \cdot \left.\frac{(16-r^2)^{3/2}}{3}\right|_2^4$

$\rightarrow 2\pi \cdot -4$

## Mathematics (2)

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2. Above the cone $z=\sqrt{x^2+y^2}$ and below the sphere $x^2+y^2+z^2=1$.

$\int$

$z=\sqrt{1-(x^2+y^2)}$ *

$z=\sqrt{1-(r)}$.

$\int_{-\infty}^{\infty} e^{-y^2}\,dy=\sqrt{\pi}$

iii) Given that $\int_{-\infty}^{\infty} e^{-x^2}dx=\sqrt{\pi}$ , evaluate

1. $\int_0^{\infty} x^2 e^{-x^2}dx$

$\cancel{I=\int e^{-x^2}dx}$

$I=\int e^{-y^2}dy$.

$\int x^2 e^{-x^2}\,dx$

$u=x \qquad dv=xe^{-x^2}$

$du=1 \qquad v=\frac{-e^{x^2}}{2}$.

$-\frac{xe^{-x^2}}{2}+\frac{1}{2}\int^{\infty} \cancel{\phantom{x}}\, e^{-x^2}$

$-\frac{\cancel{xe^{-x^2}}}{2}\Big|_{-\infty} + \frac{\sqrt{\pi}}{2} \cancel{x}\frac{1}{2} = \frac{\sqrt{\pi}}{4}$

2. $\int_0^{\infty} \sqrt{x}\, e^{-x} dx$

$\cancel{I}\, 2 \qquad y=\sqrt{x}$

$y^2=x$.

$\cancel{2y\,dy=dx}$

$\cancel{dy=\frac{dx}{2y}}=$

$\cancel{\int_0^{\infty} y\, e^{-y^2}\, 2y.\,dy.}$

$u=y \qquad dv=2$

$du=1 \qquad v=$

$-ye^{-y^2}+\int_0^{\infty} e^{-y^2}\,d$

$=\frac{\sqrt{\pi}}{2}$

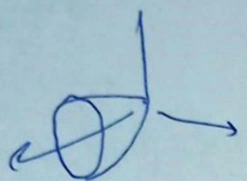
Name: Mahmood Hassan. ID: 23P0104. 

## D. Triple Integral

**1.** Use a triple integral to find the volume of the solid enclosed by the paraboloid $x = y^2 + z^2$ and the plane $x = 16$.

$\iiint dv \rightarrow \quad \iiint dA\, dx.$ $\qquad y^2 + z^2 = 16$

$$\int_0^{16}\int_0^{2\pi}\int_0^{4} r\, dr\, d\theta\, dx.$$



$$\int_0^4 r\, dr \cdot \int_0^{2\pi} d\theta \quad \int_0^{16} dx.$$

$$\boxed{256\pi}$$

**2.** Sketch the solid whose volume is given by the iterated integral.

$$\int_0^1 \int_0^{1-x} \int_0^{2-2z} dy\, dz\, dx$$

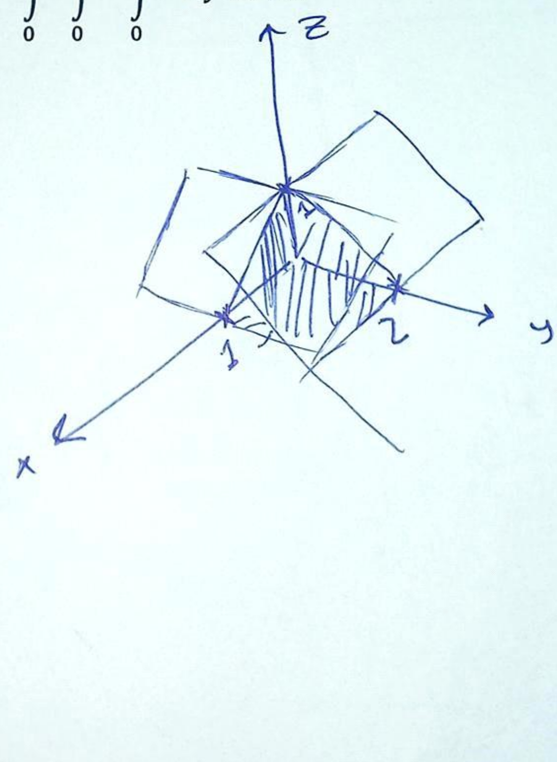
$y = 2 - 2z$

$y = 0$

$z = 1 - x$

$z = 0$

$x = 0 \quad x = 1$



Mathematics (2)

**Name:** Mahmoud Hassan **ID:** 23P0164 

3. Sketch the solid whose volume is given by the integral and evaluate the integral.

$$\int_0^{\pi/2}\int_0^{2}\int_0^{9-r^2} r\,dz\,dr\,d\theta$$

A.

$$\int_0^{\pi/2}\int_0^{2} zr\Big|_0^{9-r^2}$$

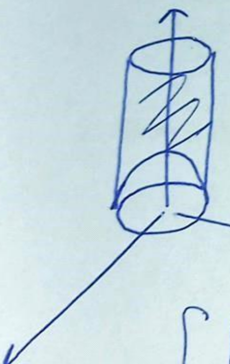
$$\int_0^{2} (9-r^3)\,\cancel{r}\,dr \cdot \int_0^{\pi/2} d\theta$$

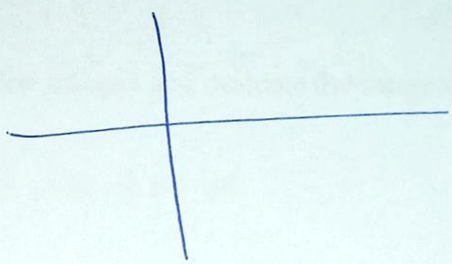
$$\frac{9}{2}r^2-\frac{r^4}{4}\Big|_0^{2} = 14\times\frac{\pi}{2} = 7\pi$$

4. Use cylindrical coordinates to

A) evaluate $\iiint_E e^z\,dV$, where $E$ is enclosed by the paraboloid $z = 1 + x^2 + y^2$, the cylinder $x^2 + y^2 = 5$ and the $xy$-plane.

A





$$\iiint_{x^2+y^2-5}^{1+x^2+y^2} e^z\,dz\,dA$$

$$\int e^{1+x^2+y^2} - e^{x^2+y^2-5} \rightarrow \int_0^{2\pi}\int_0^{\sqrt{5}} e^{1+r^2} - e^{r^2-5}$$

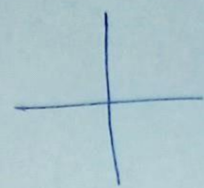
## Mathematics (2)

**Name:** Mahmoud **ID:** 23P0104. **8/12**

**B)** Find the mass of the solid $S$ bounded by the paraboloid $z = 4x^2 + 4y^2$ and the $z = a$ $(a > 0)$ if $S$ has constant density $K$.

$$\text{Mass} = \iint k \, dA$$

$$= \iint k \, r \, dr \, d\theta$$



$$z = 4x^2 + 4y^2$$
$$= 4r^2$$

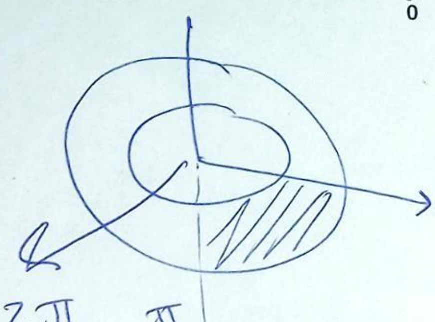
$$\int_0^{2\pi}\int_0^{a} k \, r \, dr \, d\theta$$

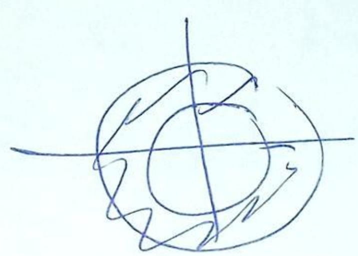
$$= \int_0^{2\pi} \left[\tfrac{1}{2} k r\right]_0^{a} d\theta = \tfrac{1}{2} k \int_0^{2\pi} a^2$$

$$= \tfrac{1}{2} k \, 2\pi \, r a^2 = 12 a^2 \pi$$

**5.** Sketch the solid whose volume is given by the integral and evaluate the integral.

$$\int_0^{2\pi}\int_{\pi/2}^{\pi}\int_1^{2} \rho^2 \sin\phi \; d\rho \; d\phi \; d\theta$$





$$\int_0^{2\pi}\int_{\pi/2}^{\pi} \frac{\rho^3}{3} \sin\phi \Big|_1^2 \, d\phi \, d\theta$$

$$\frac{7}{3} \int_{\pi/2}^{\pi} \sin\phi \, d\phi \int_0^{2\pi} d\theta = \frac{14}{3}$$

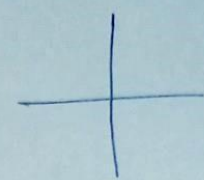
## Mathematics (2)

**Name:** Mahmoud **ID:** 23P0104. **8/12**

B) Find the mass of the solid $S$ bounded by the paraboloid $z = 4x^2 + 4y^2$ and the $z = a$ $(a > 0)$ if $S$ has constant density $K$.

$$\text{Mass} = \iint k \, dA$$

$$= \iint k \, r \, dr \, d\theta.$$



$$z = 4x^2 + 4y^2$$
$$= 4r^2$$

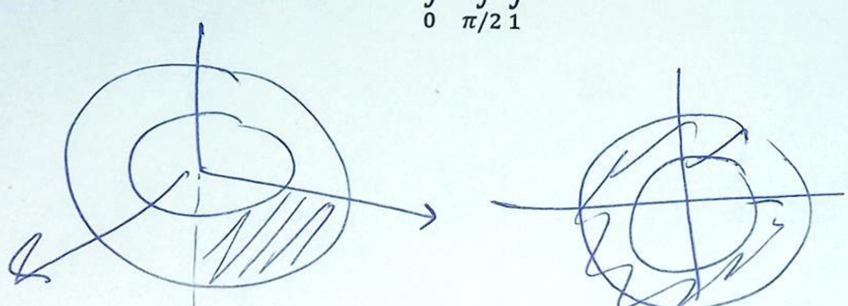
$$\int_0^{2\pi} \int_0^{a} k \, r \, dr \, d\theta.$$

$$= \int_0^{2\pi} \left[\tfrac{1}{2} k r\right]_0^a d\theta = \tfrac{1}{2} K \int_0^{2\pi} a^2 \, d\theta$$

$$= \tfrac{1}{2} k \, 2\pi \, r a^2 = 12 a^2 \pi$$

**5.** Sketch the solid whose volume is given by the integral and evaluate the integral.

$$\int_0^{2\pi} \int_{\pi/2}^{\pi} \int_1^2 \rho^2 \sin\phi \, d\rho \, d\phi \, d\theta$$



$$\int_0^{2\pi} \int_{\pi/2}^{\pi} \frac{\rho^3}{3} \sin\phi \Big|_1^2 \, d\phi \, d\theta$$

$$\frac{7}{3} \int_{\pi/2}^{\pi} \sin\phi \, d\phi \int_0^{2\pi} d\theta = \frac{14}{3}$$

## Mathematics (2)

**Name:** Mahmoud Hassan **ID:** 23P0104 **9/12**

6. Use spherical coordinates to evaluate $\iiint_E e^{\sqrt{x^2+y^2+z^2}}\, dV$, where $E$ is enclosed by the sphere $x^2+y^2+z^2=9$ in the 1st octant.

$$x^2+y^2+z^2=\rho^2$$

$$\int_0^{\pi/2}\int_0^{\pi/2}\int_0^{3} e^{\rho}\,\rho^2 \sin\phi \; d\rho\, d\phi\, d\theta$$

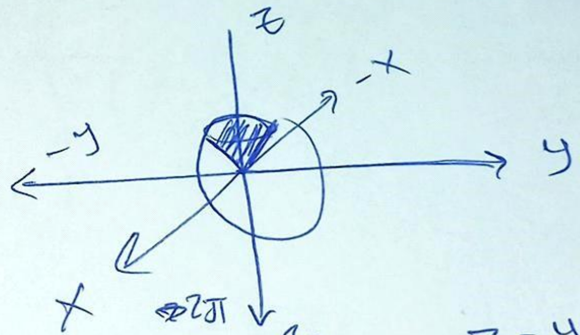
$$\left[\rho^2 e^{\rho} - 2\rho e^{\rho} + 2e^{\rho}\right]\Big|_0^3 \cdot \int_0^{\pi/2} \sin\phi\, d\phi \cdot \int_0^{\pi/2} d\theta$$

| $d$ | $\int$ |
|---|---|
| $\rho^2$ | $e^{\rho}$ |
| $2\rho$ | $e^{\rho}$ |
| $2$ | $e^{\rho}$ |
| $0$ | $e^{\rho}$ |

$$= 128.605$$

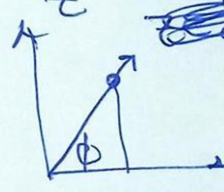
7. Use cylindrical or spherical coordinates (whichever seems more appropriate) to find the volume of solid $E$ that lies above the cone $z=\sqrt{x^2+y^2}$ and below the sphere $x^2+y^2+z^2=1$.

$$\int_0^{2\pi}\int_0^{\pi/4}\int_0^{1} \rho^2 \sin\phi \; d\rho\, d\phi\, d\theta$$




$$\int_0^1 \rho^2\, d\rho \cdot \int_0^{\pi/4} \sin\phi\, d\phi \cdot \int_0^{2\pi} d\theta$$

$z = y$




$\phi = 45$

$$\boxed{= 0.613}$$

## E. Line Integral and Green Theorem

1. Evaluate the following line integral, where $C$ a given curve.

$$\int_C x^2 y \sqrt{z}\, ds, \quad C: x = t^3,\ y = t,\ z = t^2, \qquad 0 \le t \le 1$$

$\int x^2 y \sqrt{z} \sqrt{\phantom{x}}$

$\frac{dx}{dt} = 3t^2$

$\frac{dy}{dt} = 1$

$\frac{dz}{dt} = 2t.$

$\int \cancel{t}^6 t . t \rightarrow \int t^8 \cdot \sqrt{9t^4 + 4t^2 + 1}$

$\sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt^2}\right)^2}$

$\sqrt{(3t^2)^2}$

$= 0.246.$

2. Find the arc length of the curve $y^2 = x^3$ between the points (1,1) and (4,8).

$y = x^{3/2}$

$dy = \frac{3}{2} x^{\frac{1}{2}} dx$

$\int ds$

$\int_1^4 \left(1 + \frac{9}{4}x\right)^{\frac{1}{2}} dx$

$ds = \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, dx.$

$\sqrt{1 + \left(\frac{3}{2} x^{\frac{1}{2}}\right)^2}$

$= 7.63.$

$\sqrt{1 + \frac{9}{4} x} \quad dx$